

PR1 — W001 + W002 + W003 + W004: Colour-line model, layer hierarchy & geometric tracing

Paste this as the GitHub PR description. **Do not auto-merge** — open for engineering review only.

Branch: wave/w001-w004-colour-line-layer-geometry → **Base:** main

Programme: Mott MacDonald / SCK CEN — MYRRHA-MINERVA Phase 1 (QCELL / RFCELL P&ID)

Toolchain: Python stdlib + cairosvg (PDF) + openpyxl (Excel)

What this PR delivers

A single consolidated foundation covering four waves:

- **W001** — source ingestion & style extraction (inline-style colour precedence).
- **W002** — colour-line decomposition into canonical process lines + validation.
- **W003** — 13 top-level layers (21 named sub-layers) hierarchy, per-element assignment.
- **W004** — geometric arrow/flow tracing, element pairing, component catalog, PEMO SSOT.

We deliberately do **not** open one PR per wave — the reviewer sees one coherent capability set.

Reproducibility (read first)

Derived outputs (data/model/ , data/pemo/ , output_v6/ , publish/ , reports/*.xlsx)

are **git-ignored** — regenerable, not source. From a fresh clone:

```
./make.sh                                # regenerate ALL derived
outputs
PYTHONPATH=src python3 tests/test_integration_pipeline.py # source-only smoke test
(no data needed)
PYTHONPATH=src python3 tests/test_colour_model.py        # W002 assertions (after
make.sh)
PYTHONPATH=src python3 tests/test_w003_w004.py          # W003/W004 assertions
(after make.sh)
```

Verified: fresh clone → make.sh exit 0 → 20/20 tests pass → headline numbers reproduce exactly (297 components, 982→112 unmapped, pairing median 25.35px, layer sum 1888+533=2421).

Tracked source of record: src/ , segmentation/data/*.json , configs/ , data/svg/ ,
extracted/ (source reference XLSX/SVG), tests/ , reports/*.md , docs/ .

Capability Matrix (honest “Claim ≠ Complete” accounting)

CAN — implemented and verified

Capability	Evidence
Reduce unmapped elements via legend re-classification	982 → 112 (88.6 %)
CTM-resolved geometry extraction (bbox/centroid/shape)	<code>geometry.py</code> ; 7 shape classes
Pair text → components	315 pairs
Colour-classify text nodes	533 nodes
Pair dots / heat-load triangles / arrows → lines	205 / 100 / 132
Pairing distance quality (honest caveat)	752 pairs, median 25px. 10% above 355px flagged low-confidence; max 1040px (known outlier, deferred to W005 review)
Standardize text to Consolas + 4 mm tiers	533 nodes tiered
Assign 13 top-level layers (21 named sub-layers) per element	<code>layer_assignment.json</code>
Layer sum-check balances exactly	1888 drawable + 533 text = 2421, no drops
Render layered SVG + PDF	QCELL 1834/14, RFCELL 591/12
Build interactive layer-toggle atlas	<code>layered_atlas_v6.html</code>
Trace flow arrows + junctions	132 arrows, 36 junctions
Parse vertical-letter nomenclature into tree	33 segments, 8 parents
Catalog components to Excel + HTML	297 components
Catalog spec-change dots by line	S 40, V 10, B 2, uncol. 153
Detect scope boundaries	5/5 (QM/QVB/vac/Jumper/QINFRA)
Detect handover diamonds (TP#NNN)	22
Emit PEMO YAML 1.2 SSOT	122 loops, 60 heat loads

CANNOT — blocked by source data / toolchain

Capability	Reason
Map the 112 residual magenta elements to a line	No legend swatch for that colour family in the source SVG
Confirm true 3-D pipe routing / elevations	Source is 2-D schematic; no z-data
Resolve the degenerate-transform coordinate outlier to a real position	Source transform is mathematically degenerate; mitigated via viewBox bounds

DID NOT / DEFERRED — possible, deferred to a later wave

Capability	Status / target wave
XLSX tag/instrument register reconciliation + coverage delta	W005 (next, PR2)
Recover the U line (top-left, currently black)	DEFERRED — colour re-attribution pass
Bind the 77 floating arrows to source lines	DEFERRED — nearest-line heuristic
Populate 04G_E_RED / manifold header layers	DEFERRED — reserved placeholders
Interactive layered viewer (beyond static atlas)	DEFERRED — W006, PR3
Cross-drawing reconciliation, temperature/pressure annotation	DEFERRED — later waves
CI / GitHub Actions workflow	DEFERRED — W007, PR3

Review checklist

- [] `./make.sh` reproduces all derived outputs from a clean clone.
- [] Integration smoke test passes (`tests/test_integration_pipeline.py`).
- [] 13-top-level / 21-named-sub-layer assignment matches engineering intent.
- [] 297-component catalog sanity-checks against the drawing.
- [] Handover-diamond list (22) and 5 scope boundaries confirmed.
- [] Pairing low-confidence caveat (>355px, max 1040px) understood and accepted.
- [] Documented deferrals accepted (U line, 112 magenta, 77 floating arrows).

Not merged automatically

Opened for review only. Merge after sign-off. **Next:** PR2 = W005 XLSX register reconciliation.